Steve Herrin

CONTACT INFORMATION

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SUMMARY

Staff-level engineer and leader with 15+ years of experience building scalable AI, machine learning, and data platforms in biotech and scientific computing environments. Expertise in distributed systems, MLOps, production machine learning platforms, and cloud infrastructure. Track record of leading teams and translating research systems into production infrastructure.

EXPERIENCE

Biohub (was Chan Zuckerberg Initiative), Redwood City, CA — August 2024 – present

Staff Engineer, Software and AI Research

- Architected and built framework for launching and tracking biological AI model evaluation workflows, with WandB dashboard integration, supporting multiple modeling teams and dozens of researchers
- Developed a biological AI benchmarking suite in collaboration with NVIDIA, and integrated it into model training pipelines
- Productionized and deployed multiple embedding and generative models per quarter using Kubernetes and PyTorch
- Architected and implemented a fine-grained identity and permissions (AuthZ/AuthN) system using AWS Cognito, OpenFGA, Kubernetes, and FastAPI to secure access to an AI platform hosting models, datasets, and admin tooling
- Expanded CI/CD automation with additional testing, type checking, linting, and validation pipelines to accelerate AI-assisted development workflows

Pathos AI, Chicago, IL — June 2022 – July 2024

Vice President of Engineering

- Built and led a data-focused engineering organization, growing the team from scratch to 4 engineers
- Prototyped a containerized, model-agnostic LLM-based Retrieval-Augmented Generation (RAG) platform for scientific literature search and summarization using pgvector
- Designed and deployed automated cloud-native scientific computing infrastructure using GCP, Terraform, and Nextflow for distributed multi-modal analysis workloads
- Developed a federated data catalog and management system using GCP, PostgreSQL, and Python to manage and audit access to thousands of datasets and analysis artifacts
- Developed internal automation and operational tooling using TypeScript and Deno-based Slack bots

D2G Oncology, Mountain View, CA — April 2021 – May 2022

Staff Software Engineer

- Created *in silico* simulation system for PCR amplification and DNA sequencing, supporting oligo design, quality control, and automated sequencing analysis
- Built ETL and data validation pipelines using Python, Pydantic, and PostgreSQL to standardize sequencing data ingestion and integrate with Benchling LIMS
- Developed Python SDK and API for querying and traversing large-scale biomedical knowledge graph with GraphQL and REST interfaces

23andMe, Sunnyvale, CA — January 2014 – March 2021

Senior Technical Lead Engineer (∼5 years)

- Architected and led development of a cloud-native machine learning platform using Docker and AWS ECS that supported 90% of production ML workloads
- Developed secure large-scale data access platform enabling internal and external researchers to query k-anonymized genomic datasets for more than 8 million customers using SQL, HTTP API, and web portal interfaces
- Built researcher-facing genotyping services platform, enabling data-sharing partnerships and increasing increasing revenue over 2%

- Automated genomic quality control and analysis workflows using statistical modeling and machine learning techniques in scikit-learn
- Designed a unified Python data access framework integrating MySQL, HBase, and distributed data stores, ultimately adopted for all customer data services
- Built 3 generations of distributed data pipelines using Celery, Luigi, and AWS SWF, processing petabyte-scale genomic datasets with Python, C++, and R
- Led migration of a 20k+ LOC Django application from on-prem to AWS, modernizing the stack with Python 3, React, and TypeScript

Engineering Manager (∼2 years)

- Built and managed three machine learning and data engineering teams totaling 16 engineers across senior, mid-level, and junior roles
- Led engineering-wide initiatives to standardize interviewing practices and open-source release processes across the organization

SLAC National Accelerator Lab, Menlo Park, CA May 2008 – August 2013

Research Associate

- Applied machine learning algorithms & statistical analysis to improve detector energy resolution by 25%
- Developed computer vision and reconstruction algorithms for 3D cosmic ray tracking, reducing background uncertainty by 90%
- Built web-based operational tooling and monitoring systems for a multi-institution scientific collaboration using PHP and MySQL
- Designed and programmed redundant control systems integrating 600+ heterogeneous sensors to protect $10M of cryogenic xenon
- Developed large-scale scientific data processing pipelines using Python, C++, and shell scripting to analyze terabytes of experimental data
- Coordinated distributed software and hardware development across international research teams
- Mentored graduate researchers on lab, coding, and statistical technique

Skills

Languages: Python, Rust, C, C++, SQL, TypeScript, JavaScript, Shell Scripting, R, Elm, LaTeX

Machine Learning & Data: PyTorch, scikit-learn, NumPy, SciPy, Spark, MLOps, LLMs, RAG, Data Engineering, Distributed Systems

Cloud & Infrastructure: AWS, GCP, Kubernetes, Docker, Terraform, CI/CD, MLflow, Nextflow, Celery, Linux, Git

Backend: FastAPI, Django, GraphQL, REST, PostgreSQL, HBase, Pydantic

Scientific & Domain Expertise: Bioinformatics, Computational Biology, Physics, Statistical Modeling, Scientific Computing, Radio, Analog & Digital Electronics

Education

Insight Data Science, Mountain View, CA

- Postdoctoral Fellowship

Stanford University, Stanford, CA

- Ph.D. (Physics)

Rice University, Houston, TX

- B.S. (Physics)